

Catalogue metadata repair - issues #1504 and #1516

Prepared 10 September 2026; publication package finalized 12 September 2026. Six directly supported metadata corrections are separated from one provenance proposal requiring maintainer confirmation.

Four verified volume shapes

PHerc0500P2 / 20250526151718: null -> [28096, 18209, 18209]

PHerc0500P2 / 20250528085330: null -> [15838, 9423, 9423]

PHerc0500P2 / 20250820143440: null -> [7057, 4196, 4196]

PHerc0343P / 20250521134555: null -> [5398, 5057, 5057]

Dimensions are copied from the corresponding public level-0 Zarr 0/.zarray metadata in published order. Array contents and resolution are unchanged.

Observed consumer result

Selected shape lookups: 0/4 in the captured original -> 4/4 after the direct corrections.

Selected typed parent-reference lookups: 0/3 -> 2/3 with the six direct corrections -> 3/3 only when the separate segment proposal is included.

These are metadata-consistency checks, not evidence of improved segmentation, ink detection, rendering or recovered text. Original reports: ScrollPrize/villa #1504 and #1516 by @nerln; #1516 also credits @Bullo27 for independent shape checks.

Reference integrity and exact scope

Two direct parent-type corrections

PHerc0009B volume 20250521125136: derived_from.type volume -> scan; ID remains 20250509053741.

PHerc0009B volume 20250820154339: derived_from.type volume -> scan; ID remains 20250718080859.

Each referenced ID exists as a scan in the same sample, not as a volume, and agrees with the source volume scan_id.

Separate segment proposal

Segment 20250910185200 keeps derived_from.type = volume and proposes changing the parent ID from 20250718080859 to 20250820154339, which is the segment original_volume_id. Maintainer confirmation is required; the default six-correction output leaves this segment unchanged.

Safeguards

The direct variant changes exactly four properties.shape values and two creation.derived_from.type values. The proposed variant adds one creation.derived_from.id change. The utility validates record identities, evidence hashes, current values and scan relationships; conflicts abort before output is written. A second repair application makes no changes.

A patched local snapshot is not a deployed upstream fix. Maintainers should update the authoritative metadata source or export process and regenerate the published catalogue.

Reproducibility and handoff

Verification

Twelve regression tests pass on the complete captured public catalogue. Whole-document semantic comparison verifies exactly six changes in the direct variant or seven with the segment proposal, with zero unrelated semantic changes and idempotent repeat application.

Captured source

Official metadata.json retrieved 2026-09-10T16:36:42.217859+00:00; decoded payload 15,642,382 bytes; SHA-256 2551bb522f9462d001462243baa46739442899cf14d10c1405708e0f898f667f. The evidence manifest records response hashes, headers, timestamps and four Zarr metadata sources.

Exact reproduction

```
python3 repair_catalogue.py check --evidence evidence --catalogue
evidence/metadata.original.json
```

```
python3 repair_catalogue.py verify --catalogue evidence/metadata.original.json --after
metadata.six.corrected.json
```

```
python3 repair_catalogue.py verify --catalogue evidence/metadata.original.json --after
metadata.seven.proposed.json --include-segment-proposal
```

```
python3 -m unittest -v test_repair
```

For the full historical fixture, download the v1.0.0 release archive from

https://github.com/benlucius/vesuvius-catalogue-repair/releases/download/v1.0.0/Vesuvius_Catalogue_Repair_v1.0.0.zip

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The package was prepared with OpenAI Codex assistance at the submitter request. It does not claim original discovery, maintainer acceptance, hands-on VC3D use, community adoption or a guaranteed award.